1. (a) $ix - y = i(x+iy)$ $\qquad z = x+iy$

$= iz$

fu^n of z only (not z^*)

$ix - y \equiv u + iv \Rightarrow u = -y$

$v = x$

$$\left.\begin{aligned} \frac{\partial u}{\partial x} &= 0 = \frac{\partial v}{\partial y} \\ \frac{\partial u}{\partial y} &= -1 = -\frac{\partial v}{\partial x} \end{aligned}\right\} \text{CR conditions satisfied!}$$

Moreover all partial derivation exists and continuous

Therefore $ix - y$ analytic in whole of complex plane. ©

(b) $\dfrac{x - iy}{x^2+y^2} = \dfrac{1}{z}$ $\qquad \dfrac{1}{z} = \dfrac{1}{(x+iy)}$

fu^n of z only and finite except $z=0$

$$= \frac{x-iy}{(x+iy)(x-iy)}$$

$$= \frac{x-iy}{x^2+y^2}$$

$$u = \frac{x}{x^2+y^2} \; ; \; v = -\frac{y}{x^2+y^2}$$

$$\frac{\partial u}{\partial x} = \frac{1}{x^2+y^2} - \frac{2x^2}{(x^2+y^2)^2} = \frac{x^2+y^2-2x^2}{(x^2+y^2)^2} = \frac{y^2-x^2}{(x^2+y^2)^2}$$

$$\frac{\partial v}{\partial y} = -\left[\frac{1}{x^2+y^2} - \frac{2y^2}{(x^2+y^2)^2}\right] = \frac{-(x^2+y^2)+2y^2}{(x^2+y^2)^2}$$

$$= \frac{-x^2+y^2}{(x^2+y^2)^2}$$

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$$\frac{\partial u}{\partial y} = -\frac{2xy}{(x^2+y^2)^2} \quad ; \quad \frac{\partial v}{\partial x} = \frac{2yx}{x^2+y^2}$$

We see $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$; $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$

except $x=0, y=0$

Analytic ~~except~~ in whole complex plane except $z=0$

i.e $\mathbb{C} \setminus \{0\}$

20 (a) $\dfrac{z\cos z - \sin z}{z^4}$ pole can only be at $z \to 0$

Numerator $= z\left(1 - \dfrac{z^2}{2!} + \dfrac{z^4}{4!} \cdots\right) - \left(z - \dfrac{z^3}{3!} + \dfrac{z^5}{5!} \cdots\right)$

$= \cancel{z} - \dfrac{z^3}{2!} + \dfrac{z^5}{4!} - \cancel{z} + \dfrac{z^3}{3!} - \dfrac{z^5}{5} \cdots$

$\dfrac{3-1}{6} = \dfrac{2}{6} = \dfrac{1}{3}$

$= \dfrac{z^3}{3} + O(z^5) \cdots$

~~Degree // order~~ $\text{fu}^n \sim \dfrac{z^3}{3} \dfrac{1}{z^4} + O(z) \cdots$

$\sim \dfrac{1}{3} \dfrac{1}{z}$

simple pole (order 1) $\to$ Residue $\left(\dfrac{1}{3}\right)$ ✓

Also has essential singularity at $z \to \infty$

(b) $f(z) = \dfrac{1 - z^{100}}{1 - z} = \dfrac{(1-z)(1 + z + z^2 + \cdots z^{99})}{(1-z)}$

$= 1 + z + z^2 + \cdots z^{99}$. ~~$= (1 - z^{99})$~~ $(z \neq 1)$.

For $z \neq 1$ it is just polynomial and therefore analytic

At $z = 1$, fu^n a priori not well defined

but $\lim_{z \to 1} \dfrac{1 - z^{100}}{1 - z} = 100$

$z = 1$ fu^n has removable singularity

i.e $f(z)$ analytic everywhere once $f(1) = 100$ defined!

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$$\oint dz \, \frac{z^m}{z^n} \qquad m \ \& \ n \text{ are integers}$$

choose counterclockwise unit circle around origin $\quad z = e^{i\theta} \Rightarrow dz = i e^{i\theta} d\theta.$

$$\int_0^{2\pi} i e^{i\theta} d\theta \, \frac{e^{im\theta}}{e^{in\theta}}$$

$$= i \int_0^{2\pi} d\theta \, e^{i(m-n+1)\theta}$$

$$= i \left. \frac{e^{i(m+1-n)\theta}}{i(m+1-n)} \right|_0^{2\pi} \qquad \text{when } m-n+1 \neq 0$$

$$= \frac{1}{(m+1-n)} \left[1 - 1 \right] = 0$$

if $\quad m - n + 1 = 0 \quad \Rightarrow \quad m + 1 = n$

$$= i \int_0^{2\pi} d\theta = 2\pi i$$

<u>Thus</u> $\quad \oint dz \, \frac{z^m}{z^n} = 2\pi i \, \delta_{m+1, n}.$

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3. $f(z) = \dfrac{1}{(z-1)} \dfrac{1}{(z-2)}$

expansion in $|z| > 2$ if $z = \dfrac{2}{u}$

$\Rightarrow \dfrac{2}{|u|} > 2 \qquad \Rightarrow \dfrac{u}{2} = \dfrac{1}{2}$

$\Rightarrow |u| < 1$

$$f(z) = \frac{1}{(z-2)} - \frac{1}{(z-1)}$$

$$= \frac{1}{\left(\frac{2}{u}\right)-2} - \frac{1}{(2/u)-1}$$

$$= \frac{u}{2-2u} - \frac{u}{2-u}$$

Both expansion ok!

$$= \frac{u}{2}\frac{1}{(1-u)} - \frac{u}{2}\frac{1}{\left(1-\frac{u}{2}\right)}$$

$|u| < 1$

$\Downarrow$

$\left|\frac{u}{2}\right| < 1$

$$= \frac{u}{2}\left(1+u+\frac{u^2}{4}+\cdots\right) - \frac{u}{2}\left(1+\frac{u}{2}+\frac{u^2}{4}+\cdots\right)$$

$$= \frac{1}{z}\left(1+\frac{2}{z}+\frac{4}{z^2}+\cdots\right) - \frac{1}{z}\left(1+1/z+1/z^2+\cdots\right)$$

$$= \sum_{n=0}^{\infty}\left(2^n-1\right)\frac{1}{z^{n+1}}$$

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5. (i) $$\oint \frac{dz}{(z-1)(z-2)^2}$$

$C_1 = 4e^{i\theta}$

$$= 2\pi i \left[\text{Res } f(z)\Big|_{z=1} + \text{Res } f(z)\Big|_{z=2} \right]$$

$$= 2\pi i \left[\lim_{z\to 1} \frac{1}{(z-2)^2} + \lim_{z\to 2} \frac{1}{(z-1)} \right]$$

$$= 2\pi i \left[1 + (-1) \right] = 0$$

(ii) $$\oint \frac{dz}{(z-1)(z-2)^2}$$ $z = Re^{i\theta}$

$dz = Re^{i\theta} i\,d\theta$

Circle of $R\to\infty$

$$\sim \frac{R}{R \cdot R^2} \sim \frac{1}{R^2} \xrightarrow[R\to\infty]{} 0$$

(iii) Moving ~~[illegible]~~ contour of radius 4 to ∞ does not change the value of the integration as we can distort the contour continuously. So (i) = (ii).

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c. $$\int_{-\alpha}^{\alpha} \frac{dx \quad x \sin x}{x^2 + a^2}$$

$$= \operatorname{Im} \int_{-\alpha}^{\alpha} \frac{x e^{ix}}{x^2 + a^2} dx$$

Consider $a > 0$

$$= \operatorname{Im} \left[\oint \frac{z e^{iz}}{z^2 + a^2} dz - \int_{\Gamma_{UHC}} \frac{z e^{iz}}{z^2 + a^2} dz \right]$$

$R \to \infty$ $\quad \Downarrow$ 0

Jordan's Lemma.

$\frac{1}{z + ia}$

Pole at $z = \pm ia$.

$$= \operatorname{Im} \left[\operatorname{Res} \left[z e^{iz} \frac{1}{z^2 + a^2} \right] \right]_{z = ia}$$

$$= \operatorname{Im} \left[\cancel{2\pi i} \, \cancel{i} \, e^{i(ia)} \frac{1}{\cancel{2i}\,\cancel{a}} \right]$$

$$= \operatorname{Im} \left[i\pi e^{-a} \right] = \pi e^{-a}$$

Consider $a < 0$ do explicitly by closing LHC

or note I depends on a^2

thus $$I = \pi e^{-|a|} \qquad a \neq 0$$

For $a = 0$ $$\int_{-\alpha}^{\alpha} \frac{\sin x}{x} dx = \lim_{a \to 0} \pi e^{-|a|} = \pi.$$

$$\boxed{I = \pi e^{-|a|}}$$

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7. $$f(t) = \frac{2\alpha}{\sqrt{2\pi}} \frac{1}{\alpha^2+t^2}$$

$$g(\omega) = \frac{1}{(2\pi)} \int_{-\infty}^{\infty} \frac{2\alpha\, e^{i\omega t}}{\alpha^2+t^2}\, dt$$

take $\omega > 0$ $$= \frac{1}{2\pi}\left[\oint_{UHC} \frac{2\alpha\, e^{i\omega z}}{\alpha^2+z^2}\, dz - \underbrace{\int_{R\to\infty} \frac{2\alpha\, e^{i\omega z}}{\alpha^2+z^2}\, dz}_{\substack{\downarrow\downarrow \\ 0 \\ \text{Jordon's Lemma}}}\right]$$

$$\frac{1}{(z+i\alpha)(z-i\alpha)}$$

$$= \frac{1}{\cancel{2\pi}} \cancel{2\pi} i \frac{\cancel{2\alpha}}{\cancel{2i\alpha}} e^{i\omega(i\alpha)}$$

$$= e^{-\alpha\omega}$$

take $\omega < 0$ $$g(\omega) = \frac{1}{2\pi}(-2\pi i)\frac{2\alpha}{(-2i\alpha)} e^{-i|\omega|(-i\alpha)}$$

$$= e^{-\alpha|\omega|}$$

Together $$\boxed{g(\omega) = e^{-\alpha|\omega|}}$$

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